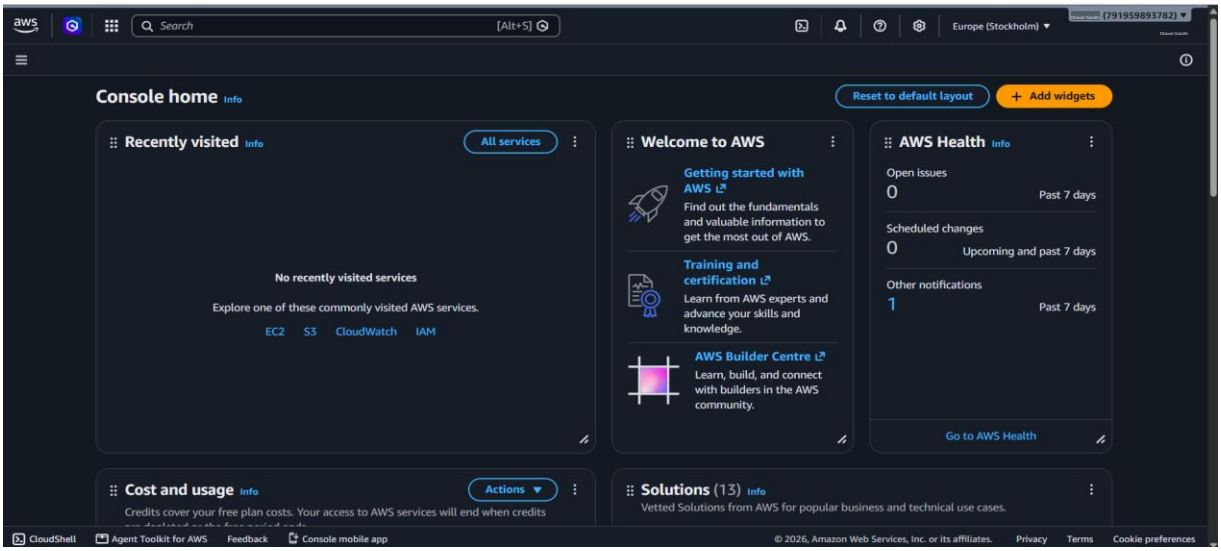
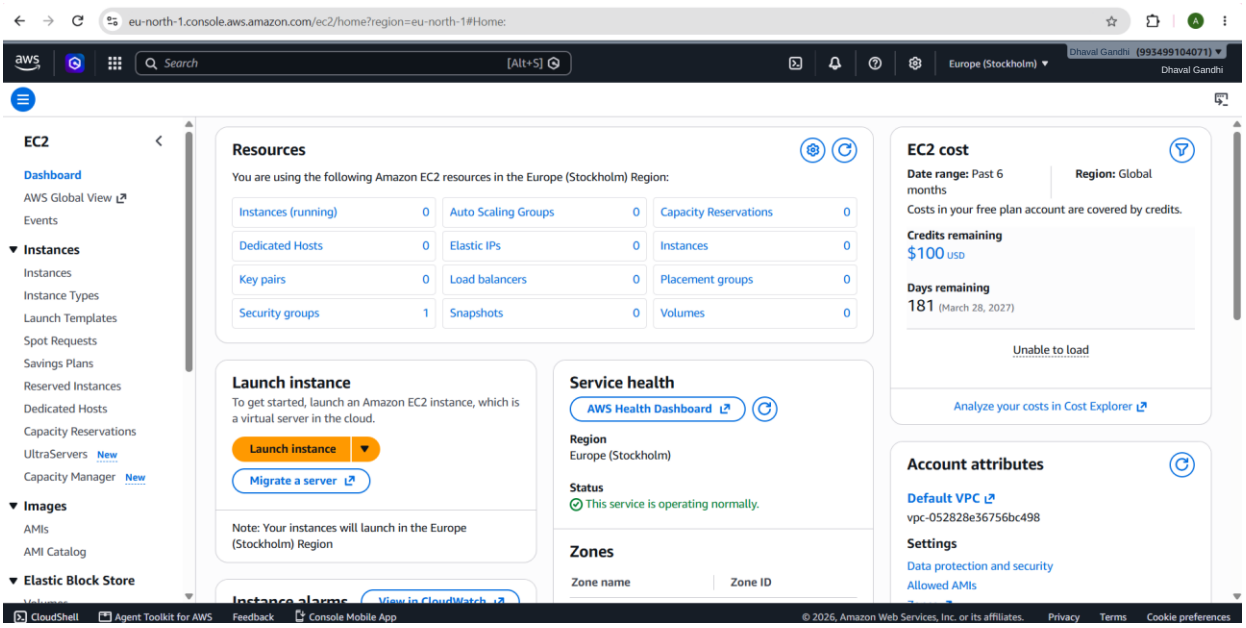
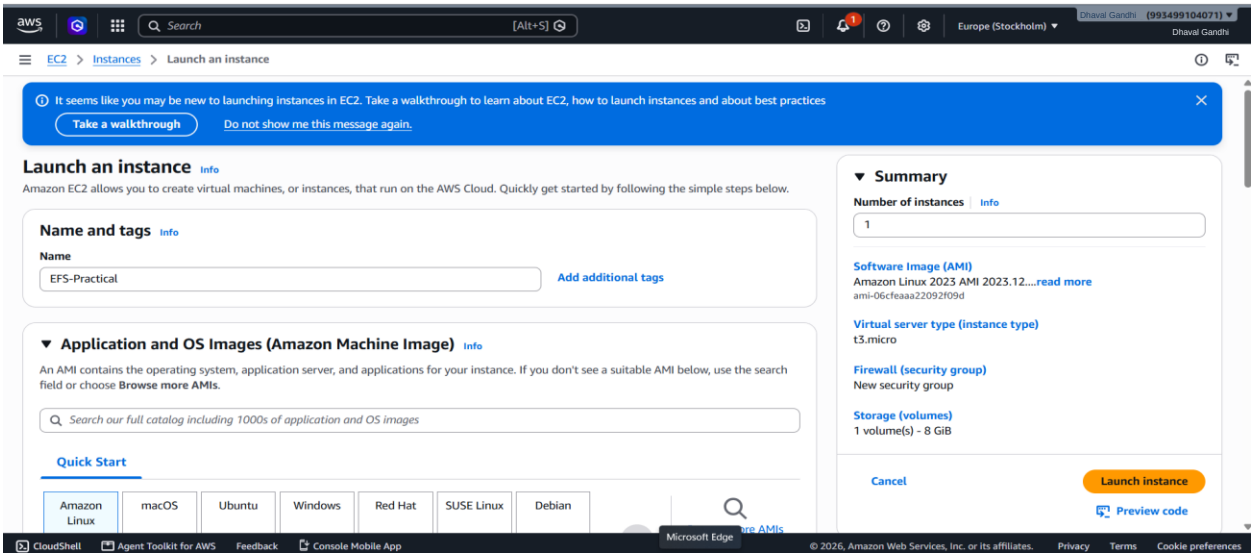
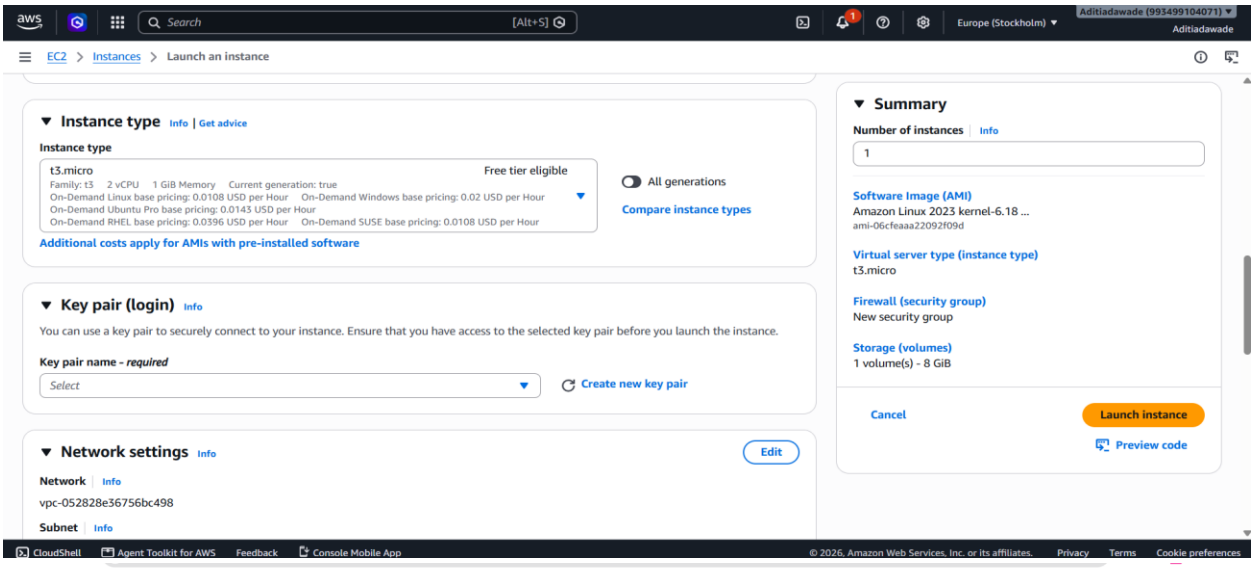
Practical no. 7

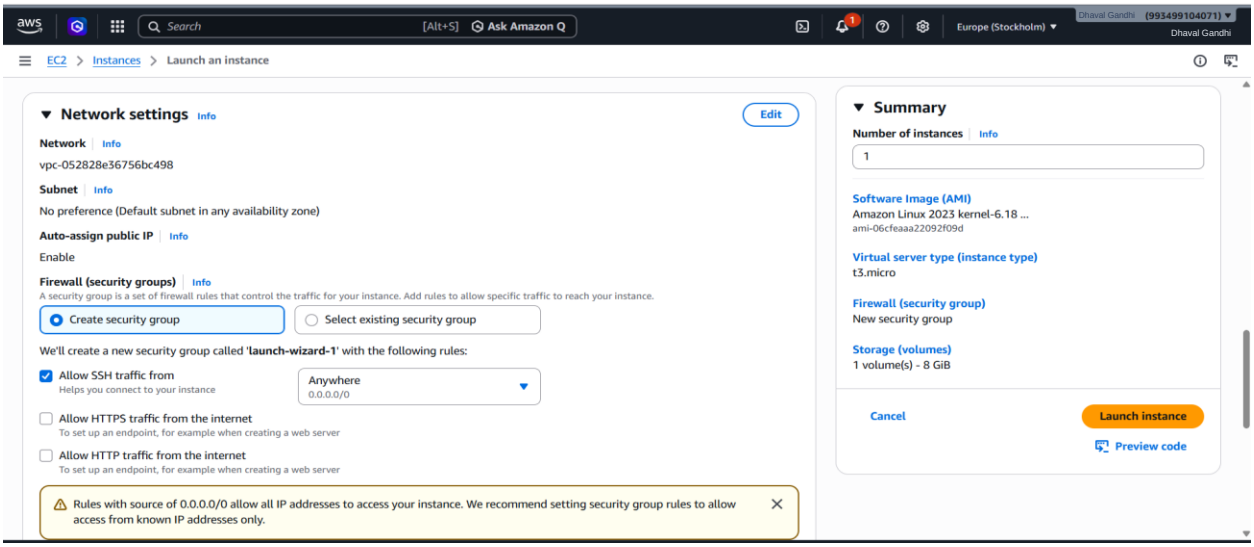
Step1: Launch an ec2 instance

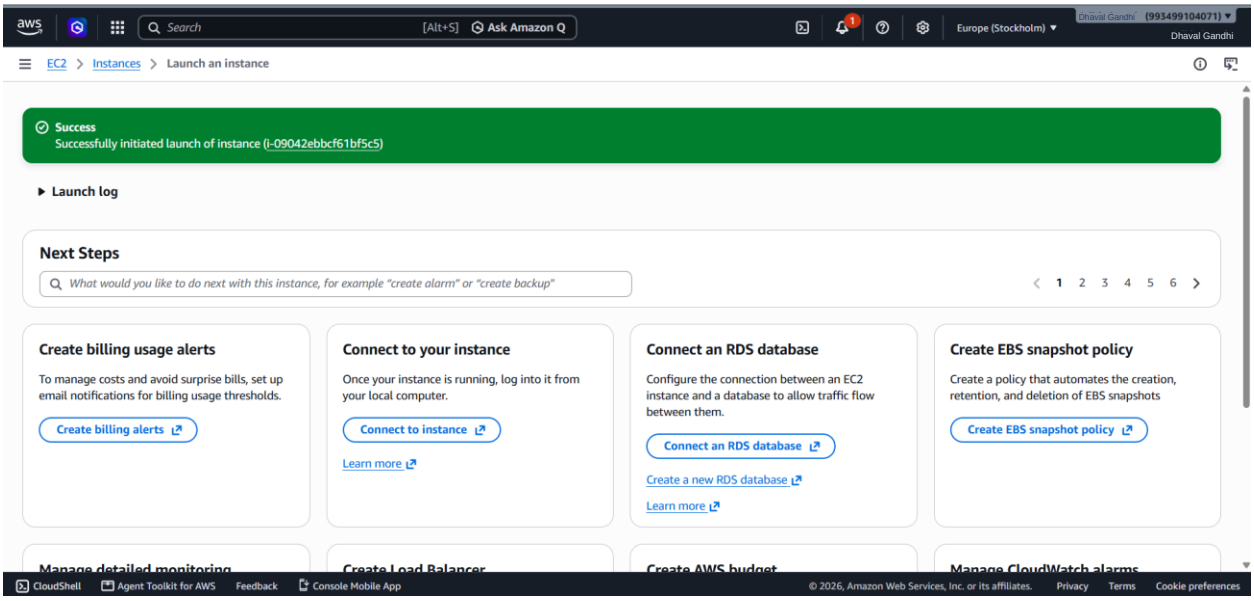




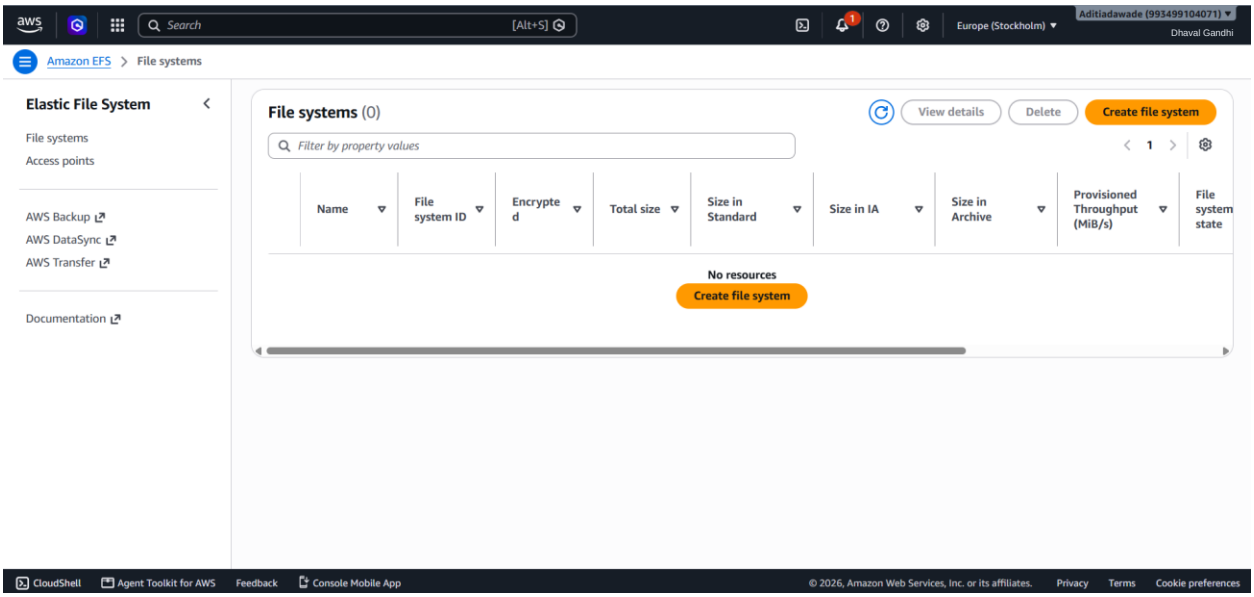

Search [Alt+S]
Europe (Stockholm)
Dhaval Gandhi (993499104071)
Dhaval Gandhi
EC2 > Instances > Launch an instance
It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices
Take a walkthrough
Do not show me this message again.
Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Name
EFS-Practical
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-06cfeaaa22092f09d
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
Microsoft Edge
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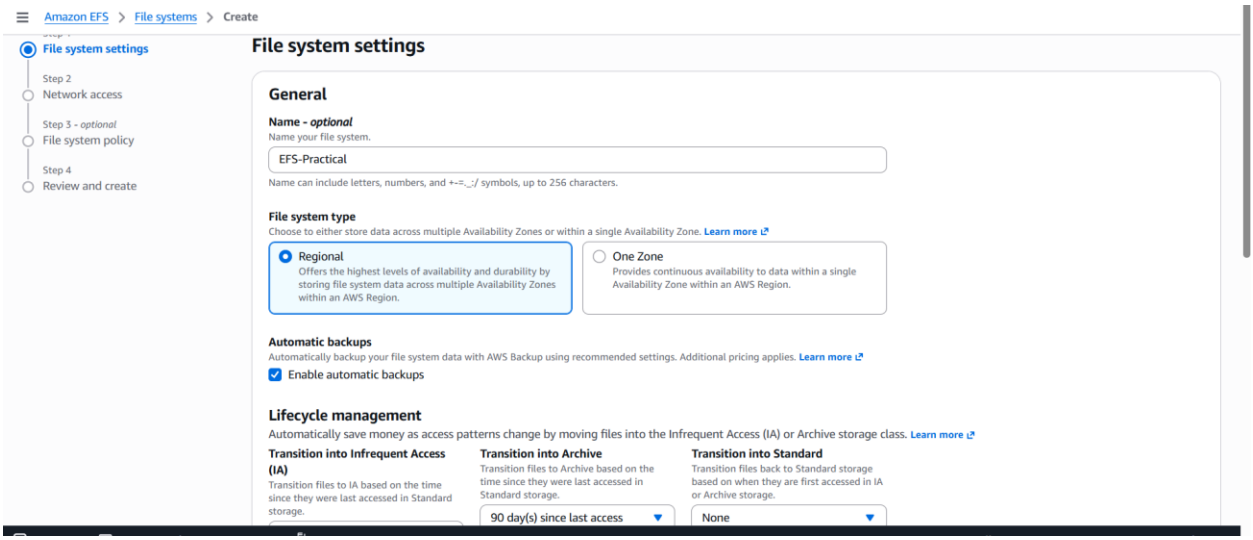

Search [Alt+S]
Europe (Stockholm)
Aditiadawade (993499104071)
Aditiadawade
EC2 > Instances > Launch an instance
Instance type Info | Get advice
Instance type
t3.micro
Free tier eligible
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0108 USD per Hour On-Demand Windows base pricing: 0.02 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0143 USD per Hour
On-Demand RHEL base pricing: 0.0396 USD per Hour On-Demand SUSE base pricing: 0.0108 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
Select
Create new key pair
Network settings Info
Edit
Network Info
vpc-052828e36756bc498
Subnet Info
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 kernel-6.18 ...
ami-06cfeaaa22092f09d
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
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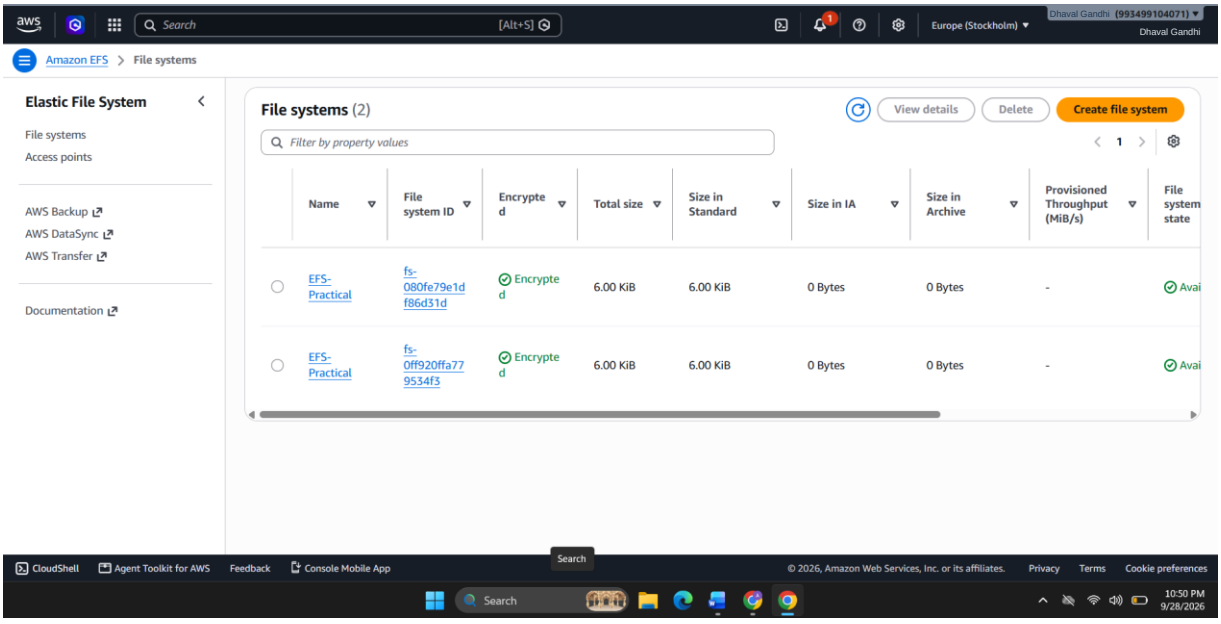

Search [Alt+S] Ask Amazon Q
Europe (Stockholm)
Dhaval Gandhi (993499104071)
Dhaval Gandhi
EC2 > Instances > Launch an instance
Network settings Info
Edit
Network Info
vpc-052828e36756bc498
Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info
Enable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
We'll create a new security group called 'launch-wizard-1' with the following rules:
Allow SSH traffic from
Helps you connect to your instance
Anywhere
0.0.0.0/0
Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server
Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server
Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 kernel-6.18 ...
ami-06cfeaaa22092f09d
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code



Step2: Create an EFS File system





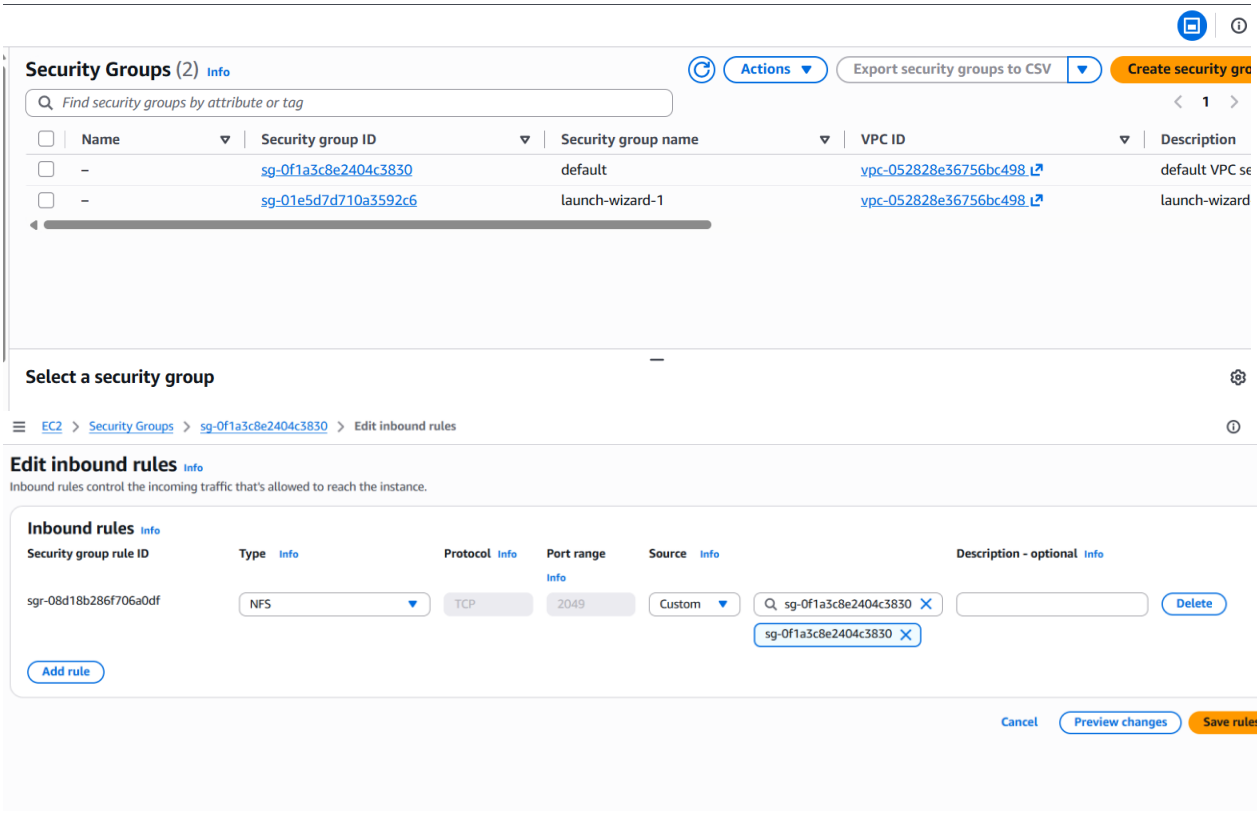


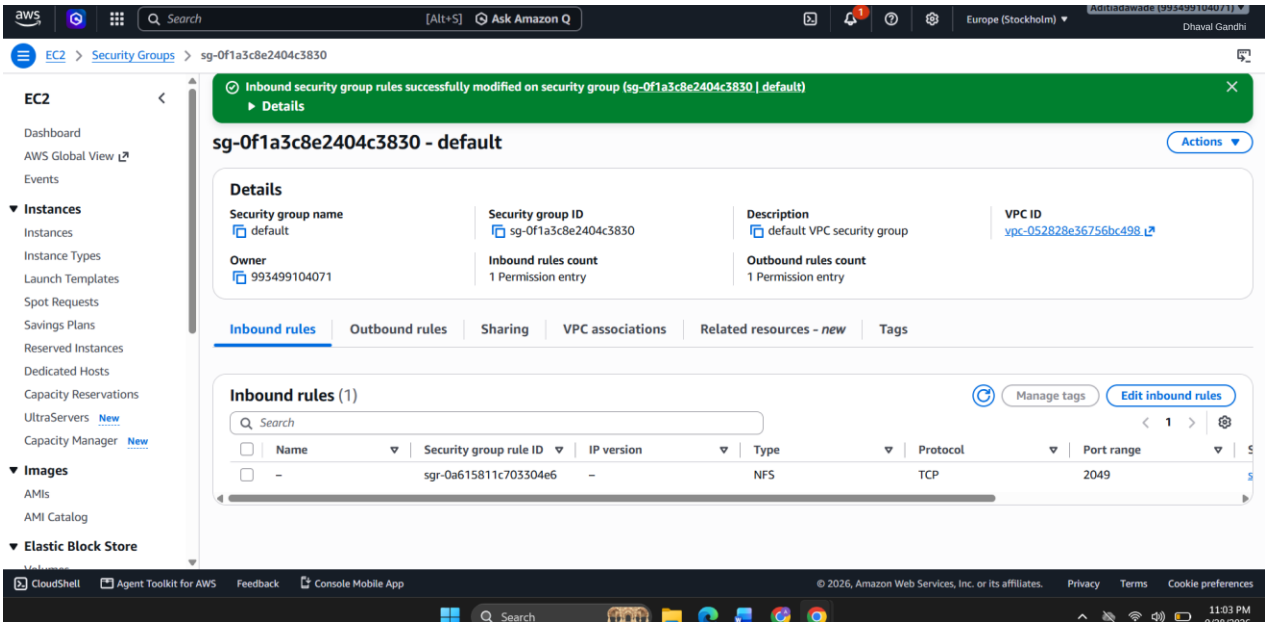
Network

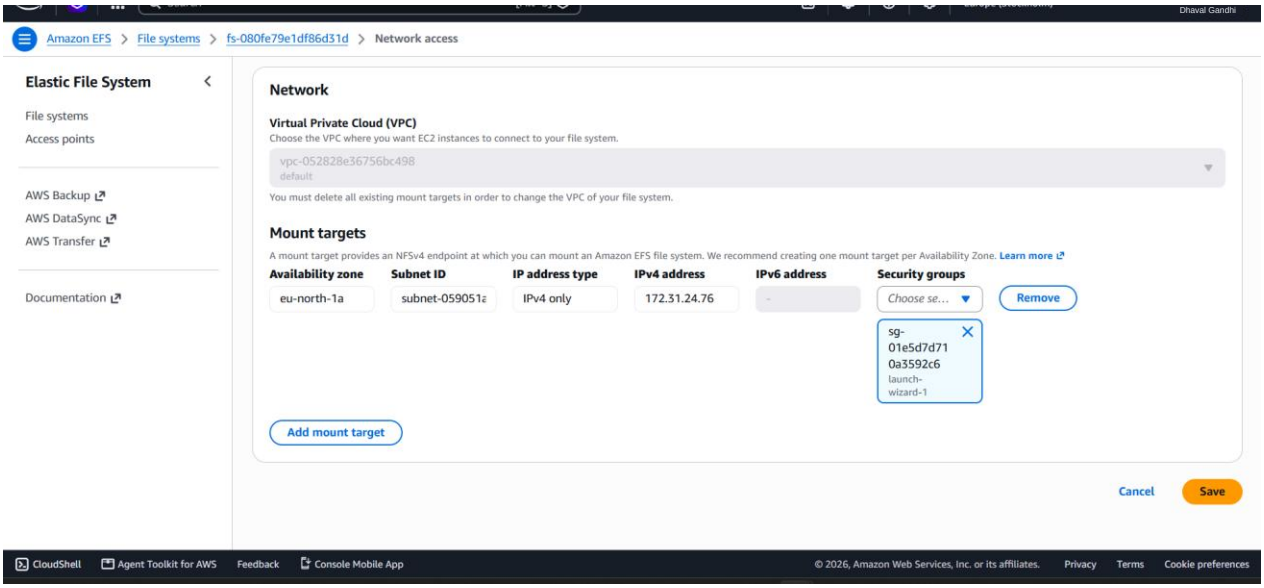
Manage

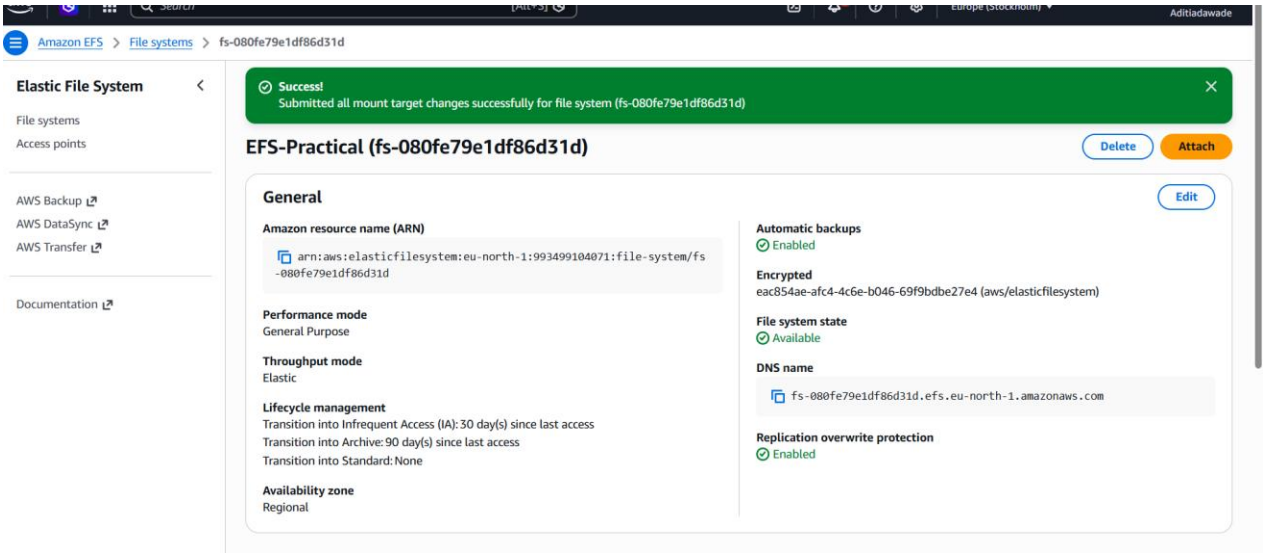
Availability zone (AZ-ID)	Mount target ID	Subnet ID	VPC ID	Mount target state	IPv4 address	IPv6 address	Network interface ID	Security groups
eu-north-1a (eun1-az1)	fsmt-0c50c08a010cf32d9	subnet-059051a422c09a2cf	vpc-052828e36756bc498	Available	172.31.24.76	N/A	eni-077e6d5c02006a2af	sg-0f1a3c8e2404c3830 (default)
eu-north-1b (eun1-az2)	fsmt-09b707cc028e892b8	subnet-0ac1df016ca5aab5a	vpc-052828e36756bc498	Available	172.31.32.231	N/A	eni-0c58312bd462a4ecf	sg-0f1a3c8e2404c3830 (default)
eu-north-1c (eun1-az3)	fsmt-0154e99040a377074	subnet-0924508f9570f599e	vpc-052828e36756bc498	Available	172.31.4.4	N/A	eni-094b4377402cda099	sg-0f1a3c8e2404c3830 (default)

Step 3:Configure Security Groups





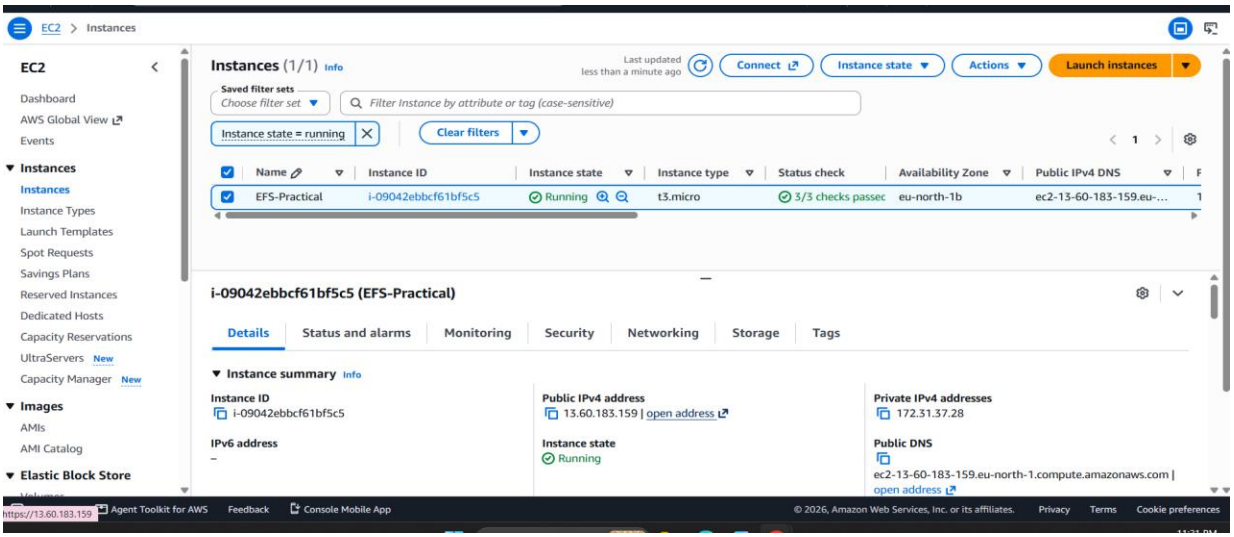


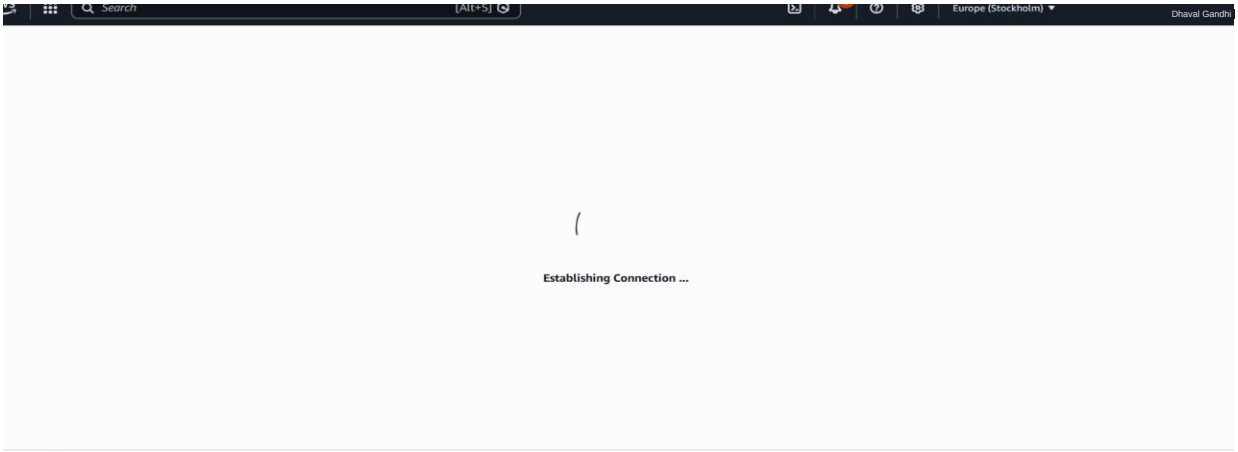


Step4: Connect to EC2 instance vis SSH

Step5: Install NFS utilites on EC2

Step6: Create a mount Directory on EC2





09042ebbcf61bf5c5 (EFS-Practical)

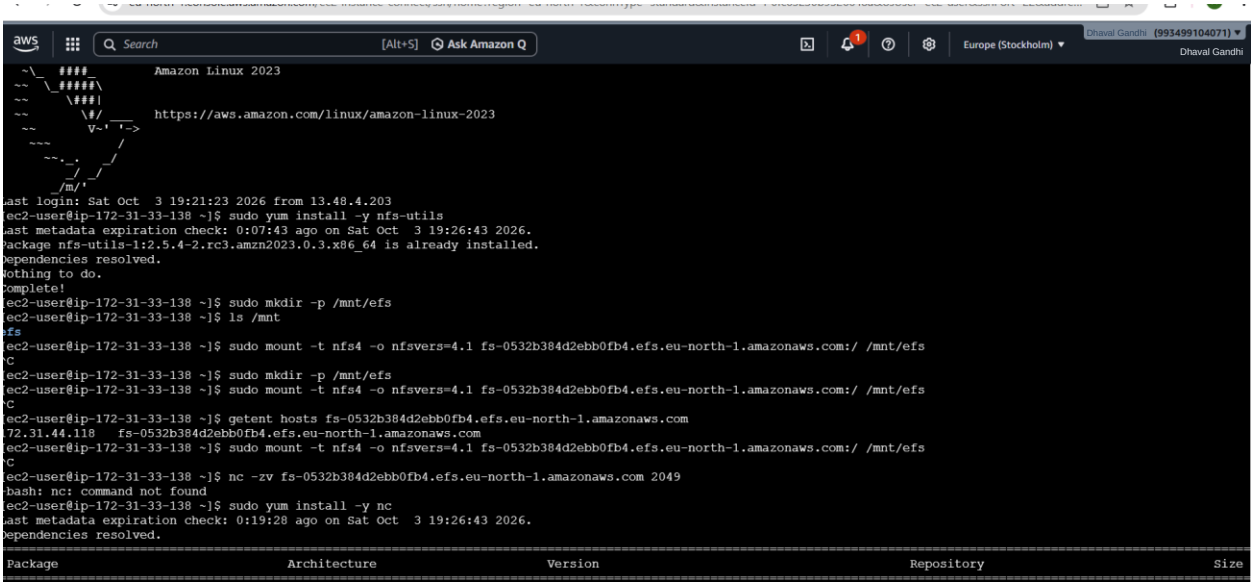
```
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___    https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-33-138 ~]$ ssh -i "your-key.pem" ec2-user@<Public-IP>
-bash: syntax error near unexpected token `newline'
[ec2-user@ip-172-31-33-138 ~]$ ssh -i "C:\path\to\your-key.pem" ec2-user@ssh51.20.3.193
Warning: Identity file C:\path\to\your-key.pem not accessible: No such file or directory.
ssh: Could not resolve hostname ssh51.20.3.193: Name or service not known
[ec2-user@ip-172-31-33-138 ~]$ sudo yum install -y nfs-utils

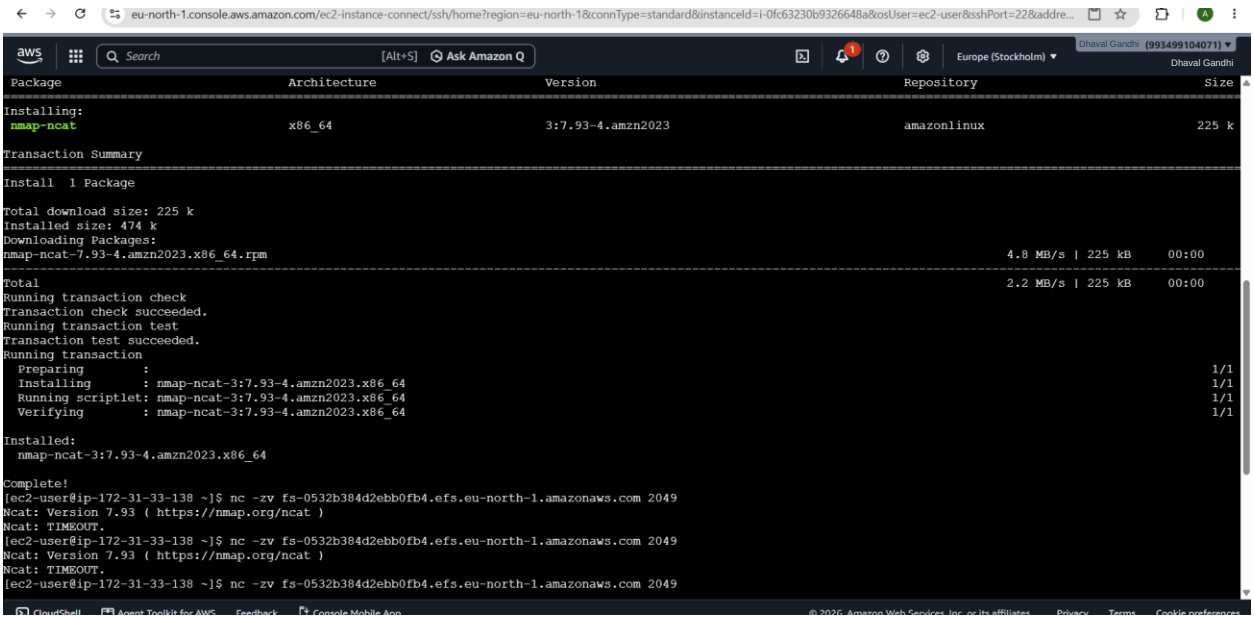
Amazon Linux 2023 Kernel Livepatch repository
Package nfs-utils-1:2.5.4-2.rc3.amzn2023.0.3.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-33-138 ~]$ sudo mkdir -p /mnt/efs
[ec2-user@ip-172-31-33-138 ~]$ ls /mnt
efs
[ec2-user@ip-172-31-33-138 ~]$
```

Step 7: Mount the EFS File System

Step 8: Verify the mount

Step 9: Test the storage by creating a file



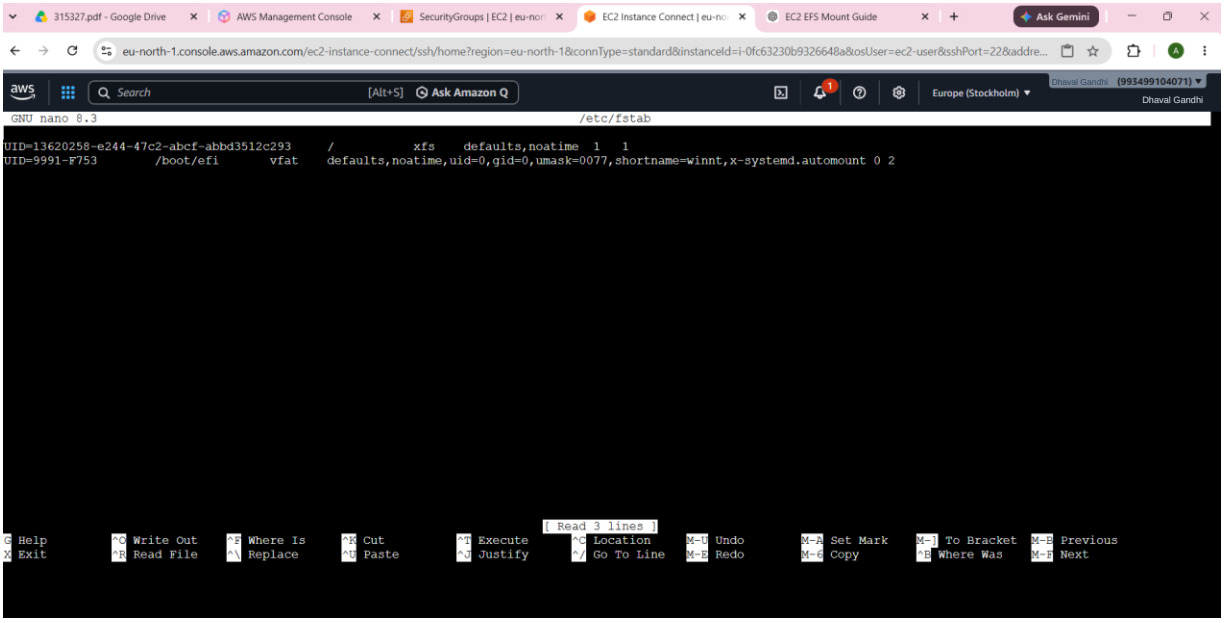


```
Ncat: Version 7.93 ( https://nmap.org/ncat )
Ncat: TIMEOUT.
[ec2-user@ip-172-31-33-138 ~]$ nc -zv fs-0532b384d2ebb0fb4.efs.eu-north-1.amazonaws.com 2049
Ncat: Version 7.93 ( https://nmap.org/ncat )
Ncat: Connected to 172.31.44.118:2049.
Ncat: 0 bytes sent, 0 bytes received in 0.01 seconds.
[ec2-user@ip-172-31-33-138 ~]$ sudo mkdir -p /mnt/efs
[ec2-user@ip-172-31-33-138 ~]$ sudo mount -t nfs4 -o nfsvers=4.1 fs-0532b384d2ebb0fb4.efs.eu-north-1.amazonaws.com:/ /mnt/efs
[ec2-user@ip-172-31-33-138 ~]$ df -h
Filesystem                                      Size  Used Avail Use% Mounted on
devtmpfs                                        4.0M     0  4.0M   0% /dev
tmpfs                                           457M     0  457M   0% /dev/shm
tmpfs                                           183M  444K  183M   1% /run
efivarfs                                        128K  2.7K  121K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p1                                  8.0G  1.7G  6.3G  21% /
tmpfs                                           457M     0  457M   0% /tmp
/dev/nvme0n1p128                                 10M  1.3M  8.7M  13% /boot/efi
tmpfs                                            92M     0   92M   0% /run/user/1000
fs-0532b384d2ebb0fb4.efs.eu-north-1.amazonaws.com:/  8.0E     0  8.0E   0% /mnt/efs
[ec2-user@ip-172-31-33-138 ~]$ echo "Hello from EFS" | sudo tee /mnt/efs/testfile.txtecho
Hello from EFS
[ec2-user@ip-172-31-33-138 ~]$ ls /mnt/efs
testfile.txtecho
[ec2-user@ip-172-31-33-138 ~]$
```

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Step 10: Make the mount persistent

Step 11: Reboot and confirm persistent



```
[ec2-user@ip-172-31-33-138 ~]$ nc -zv fs-0532b384d2ebb0fb4.efs.eu-north-1.amazonaws.com 2049
Ncat: Version 7.93 ( https://nmap.org/ncat )
Ncat: Connected to 172.31.44.118:2049.
Ncat: 0 bytes sent, 0 bytes received in 0.01 seconds.
[ec2-user@ip-172-31-33-138 ~]$ sudo mkdir -p /mnt/efs
[ec2-user@ip-172-31-33-138 ~]$ sudo mount -t nfs4 -o nfsvers=4.1 fs-0532b384d2ebb0fb4.efs.eu-north-1.amazonaws.com:/ /mnt/efs
[ec2-user@ip-172-31-33-138 ~]$ df -h
Filesystem                                      Size  Used Avail Use% Mounted on
devtmpfs                                        4.0M     0  4.0M   0% /dev
tmpfs                                           457M     0  457M   0% /dev/shm
tmpfs                                           183M  444K  183M   1% /run
efivarfs                                        128K  2.7K  121K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p1                                  8.0G  1.7G  6.3G  21% /
tmpfs                                           457M     0  457M   0% /tmp
/dev/nvme0n1p128                                 10M  1.3M  8.7M  13% /boot/efi
tmpfs                                            92M     0   92M   0% /run/user/1000
fs-0532b384d2ebb0fb4.efs.eu-north-1.amazonaws.com:/  8.0E     0  8.0E   0% /mnt/efs
[ec2-user@ip-172-31-33-138 ~]$ echo "Hello from EFS" | sudo tee /mnt/efs/testfile.txtecho
Hello from EFS
[ec2-user@ip-172-31-33-138 ~]$ ls /mnt/efs
testfile.txtecho
[ec2-user@ip-172-31-33-138 ~]$ sudo nano /etc/fstab
[ec2-user@ip-172-31-33-138 ~]$ sudo mount -a
[ec2-user@ip-172-31-33-138 ~]$ df -h
Filesystem                                      Size  Used Avail Use% Mounted on
devtmpfs                                        4.0M     0  4.0M   0% /dev
tmpfs                                           457M     0  457M   0% /dev/shm
tmpfs                                           183M  444K  183M   1% /run
efivarfs                                        128K  2.7K  121K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p1                                  8.0G  1.7G  6.3G  21% /
tmpfs                                           457M     0  457M   0% /tmp
/dev/nvme0n1p128                                 10M  1.3M  8.7M  13% /boot/efi
tmpfs                                            92M     0   92M   0% /run/user/1000
fs-0532b384d2ebb0fb4.efs.eu-north-1.amazonaws.com:/  8.0E     0  8.0E   0% /mnt/efs
[ec2-user@ip-172-31-33-138 ~]$
```

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```
tmpfs                                            92M     0   92M   0% /run/user/1000
fs-0532b384d2ebb0fb4.efs.eu-north-1.amazonaws.com:/  8.0E     0  8.0E   0% /mnt/efs
[ec2-user@ip-172-31-33-138 ~]$ sudo reboot

Broadcast message from root@localhost on pts/1 (Sat 2026-10-03 20:13:46 UTC):

The system will reboot now!

[ec2-user@ip-172-31-33-138 ~]$
```

```
       ~~._.   _/
          _/ _/
        _/m/'
Last login: Sat Oct  3 19:34:09 2026 from 13.48.4.202
[ec2-user@ip-172-31-33-138 ~]$ df -h
Filesystem                                      Size  Used Avail Use% Mounted on
devtmpfs                                        4.0M     0  4.0M   0% /dev
tmpfs                                           457M     0  457M   0% /dev/shm
tmpfs                                           183M  464K  183M   1% /run
efivarfs                                        128K  2.7K  121K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p1                                  8.0G  1.7G  6.3G  21% /
tmpfs                                           457M     0  457M   0% /tmp
/dev/nvme0n1p128                                 10M  1.3M  8.7M  13% /boot/efi
fs-0532b384d2ebb0fb4.efs.eu-north-1.amazonaws.com:/  8.0E     0  8.0E   0% /mnt/efs
tmpfs                                            92M     0   92M   0% /run/user/1000
```